ANALYSIS REPORT

Track 1 SVM Open-Cell Repair Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the dedicated `SVR` repair campaign prepared in:

- `doc/reports/campaign_plans/2026-04-14-17-17-21_track1_svm_open_cell_repair_campaign_plan_report.md`

  The campaign executed `12` exact-paper validation runs through the dedicated launcher:
- completed runs: `12`
- failed runs: `0`
- execution window: `2026-04-14 17:30:44+02:00` to `2026-04-14 17:32:30+02:00`
- campaign artifact root:

  `output/training_campaigns/track1_svm_open_cell_repair_campaign_2026_04_14_17_17_21/` The campaign goal was narrow and explicit:
- repair only the currently yellow or red `SVM` paper cells;
- prioritize the true blockers first: amplitude `156`, phase `240`, then

  `162`, then the residual bridge harmonics;
- update the canonical `SVM` row in the Track `1` benchmark with the best

  post-repair values found across the scoped runs.

Objective And Outcome

The campaign had three concrete goals

1. improve the `SVM` amplitude row without rerunning the full family matrix

   blindly;
2. improve the `SVM` phase row with special attention to the `240` blocker;
3. merge the best scoped repair values back into the canonical full-matrix

   `SVM` row. Outcome:

- all `12` runs completed successfully;
- all `12` runs exported ONNX artifacts with `0` failed exports;
- the campaign upgraded the canonical `SVM` row on all four paper surfaces;
- the `SVM` row now has `0` red cells across Tables `2-5`;
- `Track 1` remains open globally, but the `SVM` row is no longer a weak row.

  This means the campaign succeeded both operationally and scientifically as a targeted repair pass.

Ranking Policy

This campaign is not a full-matrix family comparison. Each run only covers one scoped repair slice of the `SVM` row.

The serialized campaign policy is therefore

- primary metric: `cell_closure_score_desc`
- first tie breaker: `met_paper_cell_count_desc`
- second tie breaker: `open_paper_cell_count_asc`
- third tie breaker: `mean_normalized_gap_ratio_asc`
- fourth tie breaker: `max_normalized_gap_ratio_asc`
- fifth tie breaker: `improved_cell_count_desc`
- sixth tie breaker: `status_upgrade_score_desc`
- seventh tie breaker: `run_name`

Where:

- `cell_closure_score = (met + 0.5 * near) / paper_cell_count`
- `met` means the scoped repository cell reached or beat the paper cell
- `near` means the positive gap stayed within `25%` of the paper cell
- `status_upgrade_score` rewards `red -> yellow`, `yellow -> green`, and `red -> green` transitions versus the pre-repair `SVM` baseline row Interpretation:
- this policy exists only to serialize the required campaign-best artifacts;
- the real scientific interpretation remains the merged post-repair `SVM` row across Tables `2-5`.

Campaign Ranking

Ranked Completed Runs

Rank	Run	Scope	Cells	Met	Near	Open	Closure Score	Improved	Upgrades
1	track1_svm_phase_repair_seed11	phases_only	8	6	2	0	0.875	8	7
2	track1_svm_amplitude_low_mid_bridge	amplitudes_only	6	3	3	0	0.750	4	3
3	track1_svm_phase_162_240_focus	phases_only	4	2	2	0	0.750	4	4
4	track1_svm_amplitude_repair_seed23	amplitudes_only	10	4	6	0	0.700	8	5
5	track1_svm_phase_repair_seed23	phases_only	8	2	6	0	0.625	5	4
6	track1_svm_phase_1_39_bridge	phases_only	4	0	4	0	0.500	3	0
7	track1_svm_amplitude_repair_seed11	amplitudes_only	10	3	3	4	0.450	4	3
8	track1_svm_amplitude_repair_baseline	amplitudes_only	10	0	9	1	0.450	0	0
9	track1_svm_phase_repair_baseline	phases_only	8	1	5	2	0.438	0	0
10	track1_svm_amplitude_156_focus	amplitudes_only	2	0	1	1	0.250	0	0
11	track1_svm_phase_240_focus	phases_only	2	0	0	2	0.000	0	0
12	track1_svm_amplitude_156_240_focus	amplitudes_only	4	0	0	4	0.000	0	0

Export Surface

Completed Runs	Exported ONNX Files	Failed Exports	Surrogate Exports
12	38	0	8

All `8` surrogate exports came from amplitude-side `SVR` targets that still needed the existing constant-linear-regression safeguard.

Campaign Best Run

The explicit bookkeeping winner is

- `track1_svm_phase_repair_seed11`

  It was selected because it achieved:

- the highest campaign scoped cell-closure score: `0.875`
- `6` met cells and `2` near cells out of `8`
- `0` open cells on its scoped phase subset
- the strongest blocker resolution package, including the `240` phase repair

  Important interpretation:

- this run is the campaign bookkeeping winner;
- it is **not** the whole scientific result by itself;
- the scientific output is the merged post-repair `SVM` row now written into

  the canonical benchmark.

SVM Row Impact

Table-Level Before Vs After

Surface	Before	After
Table `2` amplitude `MAE`	`5` green, `5` yellow, `0` red	`8` green, `2` yellow, `0` red
Table `3` amplitude `RMSE`	`5` green, `4` yellow, `1` red	`7` green, `3` yellow, `0` red
Table `4` phase `MAE`	`6` green, `2` yellow, `1` red	`8` green, `1` yellow, `0` red
Table `5` phase `RMSE`	`5` green, `3` yellow, `1` red	`8` green, `1` yellow, `0` red

Closed Or Upgraded Cells

- Table `2` amplitude `MAE` : `0` , `81` , and `240` moved from yellow to green
- Table `3` amplitude `RMSE` : `0` and `81` moved from yellow to green; `156`

  moved from red to yellow

- Table `4` phase `MAE` : `1` and `240` moved from yellow/red to green
- Table `5` phase `RMSE` : `1` , `39` , and `240` moved from yellow/red to green

Residual SVM Yellow Cells

- Table `2` : `40` , `156`
- Table `3` : `40` , `156` , `240`
- Table `4` : `162`
- Table `5` : `162`

  There are no remaining red `SVM` cells in Tables `2-5` .

Canonical Benchmark Impact

After merging the best repaired `SVM` cells into the canonical benchmark:

- Table `2` full-matrix totals moved from `50 / 40 / 10` to `53 / 37 / 10`
- Table `3` full-matrix totals moved from `50 / 36 / 14` to `52 / 35 / 13`
- Table `4` full-matrix totals moved from `50 / 38 / 2` to `52 / 37 / 1`
- Table `5` full-matrix totals moved from `40 / 43 / 7` to `43 / 41 / 6`

On the canonical best-family Track `1` surface, the repair campaign also closed one previously open harmonic target:

- Table `3` harmonic `0` moved from yellow to green

That update also changes the harmonic-closure reading in Table `6`:

- `1/10` harmonics now read as fully matched
- `8/10` remain partially matched
- `1/10` remains not yet matched

Main Conclusions

The campaign supports five conclusions.

1. The SVM Row Is Now Scientifically Usable

Before this campaign, the `SVM` row still contained hard red blockers on the amplitude and phase surfaces.

After the repair pass, the row has no remaining red cells.

2. Phase Repair Was The Strongest Lever

`track1_svm_phase_repair_seed11` was the cleanest scoped run and removed the `240` blocker that had been dragging both phase tables.

3. Amplitude Repair Worked, But 156 And 240 Still Need Finishing

The amplitude side improved materially, especially on `0`, `81`, and `240`.

The only remaining amplitude drag is now the yellow band around `40`, `156`, and `240`.

4. The Campaign Improved The Canonical Benchmark, Not Just Side Artifacts

This was not a local experiment that produced isolated better logs.

The best repaired cells have been promoted into the canonical `Track 1` benchmark surface.

5. The Next Repair Wave Should Leave SVM And Move To Other Rows

`SVM` is no longer the right place to spend the largest repair budget.

The next efficient closure step is to attack the remaining red cells in the other families, while keeping only a very small follow-up budget for the residual `SVM` yellow cells.

Produced Campaign Artifacts

The campaign output folder now contains the required serialized winner artifacts:

- `output/training_campaigns/track1_svm_open_cell_repair_campaign_2026_04_14_17_17_21/campaign_leaderboard.yaml`
- `output/training_campaigns/track1_svm_open_cell_repair_campaign_2026_04_14_17_17_21/campaign_best_run.yaml`
- `output/training_campaigns/track1_svm_open_cell_repair_campaign_2026_04_14_17_17_21/campaign_best_run.md`

The canonical reports updated by this closure are:

- `doc/reports/analysis/RCIM Paper Reference Benchmark.md`
- `doc/reports/analysis/Training Results Master Summary.md`